CHEMICAL METHOD FOR METAL EXTRACTION FROM PRINTED CIRCUIT BOARDS

A proposed hydrometallurgical workflow for recovering valuable metals from e-waste

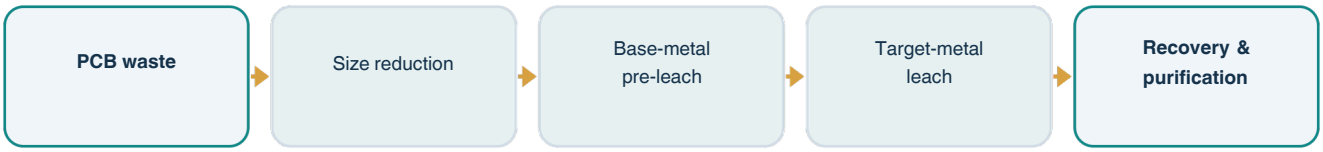
Field: Metallurgy / Hydrometallurgical Recovery of Metals
Indicative IPC class: C22B 7/00
Roll number listed in source: 24102C0025

ABSTRACT

This document describes a proposed chemical process for recovering valuable metals from waste printed circuit boards (PCBs). The workflow is organised into four stages: mechanical pretreatment, selective pre-leaching of base metals, target-metal leaching, and recovery/purification. The intended outcome is to separate metal-bearing fractions and recover selected metals for potential reuse. Actual recovery, selectivity, purity, reagent consumption and environmental performance must be established through controlled testing.

PROCESS AT A GLANCE

Four-stage process

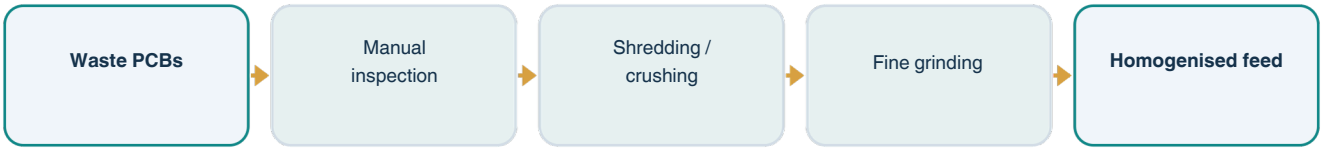


Scope note: This is a technical concept document, not a validated operating procedure or a determination that the process is patentable. Acid leaching can generate toxic fumes and hazardous waste; laboratory work requires trained supervision, suitable engineering controls and waste treatment.

1. STAGE 1 — MECHANICAL PRETREATMENT

Objective: reduce thesizeofwastePCBsandexposeembeddedmaterialssothatlaterseparation and leaching steps can act more consistently.

Pretreatment sequence



Process description

Collected PCBs are inspected and unsuitable items are removed. The boards are then reduced to smaller fragments using mechanical size-reduction equipment. Further grinding produces a more uniform feed fraction. Smaller particle size can increase exposed surface area, although excessive grinding may increase dust generation and complicate downstream separation.

Key considerations

- Remove batteries and other components that require separate handling before size reduction.
- Control dust and prevent worker exposure to fine particulate matter.
- Use suitable screening to obtain a consistent particle-size range.
- Record the mass of feed and the size fraction selected for the next stage.

Figure 1. Refined process schematic for Stage 1: pretreatment and size reduction.

Expected output: a prepared PCB-rich feed suitable for controlled downstream processing.

2. STAGE 2 — SELECTIVE PRE-LEACHING

Objective: removea portion of basemetalsbeforethetarget-metalrecoverystage,where the selected chemistry and operating conditions permit this separation.

Pre-leaching sequence

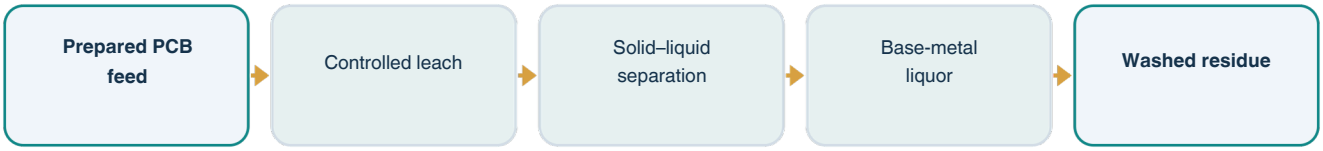


Figure 2. Refined process schematic for Stage 2: pre-leaching and separation of liquid and solid streams.

Process description

The prepared feed is contacted with a selected leaching solution in a suitable reactor. Under validated conditions, some base metals may dissolve into the liquid phase while a portion of precious-metal-bearing material remains in the solid residue. The slurry is then separated into a metal-containing solution and a solid residue. Both streams require analysis before further processing.

Important clarification

Selectivity depends on the PCB composition, reagent chemistry, concentration, temperature, contact time and particle size. It should not be assumed that copper, lead and tin will all dissolve while gold and silver remain completely undissolved. This behaviour must be confirmed experimentally using mass-balance and solution analysis.

Key considerations

- Use closed, compatible equipment and a documented risk assessment.
- Measure dissolved metals in the liquid and residual metals in the solids.
- Treat the leachate and rinse water as potentially hazardous until characterised.

3. STAGE 3 — TARGET-METAL LEACHING

Objective:transferthetargetmetalsremaininginthepre-leachedresidueintoaliquid phase for subsequent recovery.

Target-metal leaching sequence

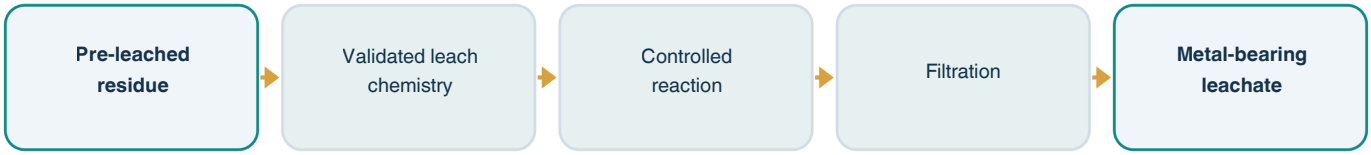


Figure 3. Refined process schematic for Stage 3: target-metal leaching followed by filtration.

Process description

The residue from Stage 2 is treated using a leaching system selected for the target metals and the measured composition of the feed. The process may require oxidation and careful control of solution chemistry. After leaching, filtration separates the liquid containing dissolved metal species from insoluble solids. The clarified solution is analysed before recovery is selected.

Chemistry and accuracy note

The original draft included reaction equations for copper, silver and gold. Some of those equations are incomplete or chemically unbalanced, and the statement that a generic acid mixture will dissolve both gold and silver is too broad. In particular, nitric acid alone does not normally dissolve gold; gold dissolution generally requires an appropriate oxidising chloride system. Reaction equations should be verified and balanced for the exact reagent system before inclusion in a formal patent or laboratory document.

Safety and process control

Oxidising acidsand acid mixtures can release highly toxic gases and cause severe chemical burns. This stage must not be treated as a do-it-yourself procedure. It requires trained personnel, a properly engineered fume-control system, compatible equipment, emergency planning and compliant hazardous-waste handling.

4. STAGE 4 — RECOVERY AND PURIFICATION

Objective: recover dissolvedmetalsfromtheclarifiedleachateandimproveproductpurityusing a separation method appropriate to the solution chemistry.

Recovery sequence



Figure 4. Refined process schematic for Stage 4: recovery, purification and product verification.

Process description

The metal-bearing leachate is analysed to determine which metal species are present and at what concentrations. A suitable recovery method—such as selective precipitation, cementation, solvent extraction followed by stripping, or electrowinning—can then be chosen. The recovered solids or metal products are separated, washed where appropriate, dried and assayed to determine composition and purity.

Important clarification

A reducing agent does not automatically precipitate gold first and silver second under all conditions. Selective recovery depends on solution composition, oxidation state, chloride concentration, pH and reagent choice. The sequence and purity claims should be supported by experimental data, analytical results and a documented material balance.

Quality checks

- Measure metal concentrations before and after recovery.
- Report recovery yield and product purity separately.
- Characterise residual solids and treatment effluent.
- Record reagent consumption and waste generated.

5. PROCESS SUMMARY AND VALIDATION PLAN

Theproposedworkflowseparatestheoperationintofourstagessothateachstreamcanbemonitored and assessed.

Stage	Main purpose	Output to verify
1. Pretreatment	Reduce size and prepare feed	Particle-size distribution and feed mass
2. Pre-leaching	Remove selected base metals	Leachate composition and residue assay
3. Target-metal leaching	Dissolve target metals from residue	Metal concentration in clarified leachate
4. Recovery / purification	Recover metals and assess purity	Recovered mass, purity and residual metals

Recommended validation work

1. Characterise the starting PCB feed and determine the initial metal inventory.
2. Conduct controlled bench-scale tests under an approved laboratory risk assessment.
3. Measure metal concentrations in every liquid and solid stream.
4. Calculate recovery yield, selectivity, product purity and material-balance closure.
5. Assess wastewater, air-emission and residue treatment requirements.
6. Compare the results with established recycling routes before making efficiency or environmental claims.

Conclusion

This document presents a staged concept for recovering valuable metals from waste PCBs through mechanical preparation, leaching and selective recovery. Its effectiveness and any claim of novelty require evidence from reproducible experiments, balanced chemistry, analytical verification and a review of relevant prior art. No specific recovery percentage or high-purity outcome is asserted without supporting data.

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